MODULARRAGOPTIMIZATION Â· GLOBAL CORRELATION EXPERIMENT

Do LLM Relevance Scores Reflect Embedding-Space Proximity to Ground Truth?

Run completed
2026-08-08 Â· 00:17 UTC

Scored pairs
1,000,174

Queries
199 total Â· 187 with GT

Scoring model
gcp/google/gemini-2.5-flash-lite Â· NVIDIA IH

Pool per query
5,026 docs (global fixed pool S)

Run duration
~3.5 days (80 RPM)

Â§ 1

Executive Summary

Central hypothesis: if an LLM scores documents for relevance to a query, do its scores reflect how close each document is â€” in embedding space â€” to the ground-truth (GT) document? A negative correlation between distance-to-GT and LLM score would validate the LLM as a geometry-aware ranker.

The answer is a strong *yes* at the population level. The LLM almost never misses GT (99.5% recovery), consistently outperforms the retriever, and is correlated with embedding proximity across all 199 queries. The main limitation is score saturation at 1.0 â€” the model cannot distinguish GT uniquely within the top tier.

SPEARMAN R (MEDIAN)

â^'0.290

100% of queries negative, 100% significant p<0.05

PEARSON R (MEDIAN)

â^'0.503

Stronger linear signal driven by GT outliers

ROC-AUC

0.9974

Near-perfect binary GT/non-GT discrimination

LLM GT RECOVERY

99.5%

186/187 queries â€” GT reaches score-1.0 tier

RETRIEVER ACCURACY

73.8%

138/187 queries retrieved GT Â· 26.2% failure rate

UNIQUE TOP SCORER

0/187

GT never uniquely rank-1 â€” median 18 docs tied at 1.0

Bottom line: The LLM is a strong coarse filter that reliably elevates GT into the top score tier. It fully recovers all 49 retriever failures (26.2% of queries) where the retriever missed GT entirely. The critical limitation is score saturation â€” the model assigns 1.0 to GT and ~17 other docs per query on average, preventing single-document selection without a secondary ranking step.

Â§ 2

Methodology & Pipeline

EXPERIMENTAL DESIGN

The experiment uses a **global fixed pool S**: the same 5,026 documents are scored against every query. This eliminates per-query pool variation and enables population-

level conclusions. Pool S was constructed by position-by-position interleaving of top-ranked docs across all 199 queries, then truncated so |S| Ã— 199 â‰ˆ 1,000,000 pairs.

Each pair records: LLM score, cosine distance from doc embedding to GT embedding (`dist_to_gt = 1 â^' cos(doc, GT)`), whether the doc is the GT (`is_gt`), and whether the retriever failed to surface GT (`rag_failed`).

0 **Embedding + Initial Ranking**

EC2 GPU ran `embed_and_rank.py`. Output: `scoring_queue.parquet` â€" 39,800 rows (199 queries Ã— top-200 docs).

1 **Global Pool Construction**

`build_cross_product.py` built pool S (5,026 docs via interleaving). Expanded to 1,000,174 pairs â†' `expanded_scoring_queue.parquet` (806 MB).

2 **LLM Scoring**

`score_from_queue.py` â†' `gcp/google/gemini-2.5-flash-lite` via NVIDIA IH (80 RPM). Non-blocking dispatcher, exponential backoff 10â†'120 s, resume-safe via SQLite NULLs.

3 **Analysis**

`analyze_experiment.py`: per-query correlation, ROC-AUC, MRR, P@K, distance quantiles, score distributions across all 1,000,174 pairs.

DATASET	**MedRAG / PubMed**	TOTAL PAIRS	**1,000,174**
POOL |S|	**5,026 docs**	QUERIES	**199 (187 with GT)**
RETRIEVER FAILURES	**49 / 187**		

Â§ 3

Component Accuracy: LLM vs. Retriever

Each of the 187 queries with a known GT document is classified by whether the retriever surfaced GT in its initial ranking, and whether the LLM placed GT at strict rank 1 (score strictly higher than all non-GT docs).

2Ã—2 BREAKDOWN (N = 187 QUERIES)

	LLM RANKED GT #1	LLM MISSED GT #1
RETRIEVER FOUND GT	**137** 73.3% of queries Both correct âœ“	**1** 0.5% â€” only query 29 Retriever only
RETRIEVER MISSED GT	**49** 26.2% â€” LLM recovers all retriever failures âœ“	**0** 0.0% â€” neither fails simultaneously

Key finding: The LLM fully compensates for all 49 retriever failures â€” in every case where the retriever missed GT, the LLM still placed GT at rank 1. Only 1 query failed the LLM alone (query 29, where GT scored 0.2 vs. competitors' 1.0). No query failed both components simultaneously.

INDIVIDUAL COMPONENT RATES

COMPONENT	SUCCESS	FAILURE	SUCCESS RATE
LLM (strict rank 1)	186 / 187	1 / 187	99.5%
Retriever (GT in pool)	138 / 187	49 / 187	73.8%
Combined (either correct)	187 / 187	0 / 187	100.0%
Both correct simultaneously	137 / 187	50 / 187	73.3%

Note on "success" definition: LLM success here = *strict rank 1* (GT score > all non-GT scores). If success is defined as GT reaching the score-1.0 tier (optimistic), LLM success = 186/187 = 99.5% and tie-corrected remains 1/187 = 0.5%. The single failure (query 29) has GT score = 0.2 while competitors score 1.0 â€” a true model error, not a tie.

Score Saturation & Tie Analysis

In 186 of 187 queries, the GT document receives score 1.0. The problem: multiple non-GT documents also receive 1.0. Understanding the tie pool size is critical for assessing whether the LLM can function as a final ranker.

DOCS TIED AT SCORE 1.0 PER QUERY

METRIC	VALUE	IMPLICATION
GT score = 1.0	186 / 187 (99.5%)	LLM almost always scores GT as maximally relevant
GT unique top scorer (ties = 1)	0 / 187 (0.0%)	GT is never the sole top scorer â€” always tied
Median tied docs at GT score	18	On a typical query, 18 docs share the top score
Mean tied docs at GT score	18.9	â€”
Min / Max tied docs	2 / 38	Even the best-case has 2 tied docs
Queries with < 5 tied docs	5 / 187 (2.7%)	Small subset with tightest discrimination
Queries with < 10 tied docs	30 / 187 (16%)	16% could use a simple re-rank
Queries with < 20 tied docs	135 / 187 (72%)	Majority manageable with secondary ranker

TOP-5 SCORING DOCUMENTS VS. GT SCORE

For each query, the 5 highest-scoring documents (including GT) and whether they match GT's score:

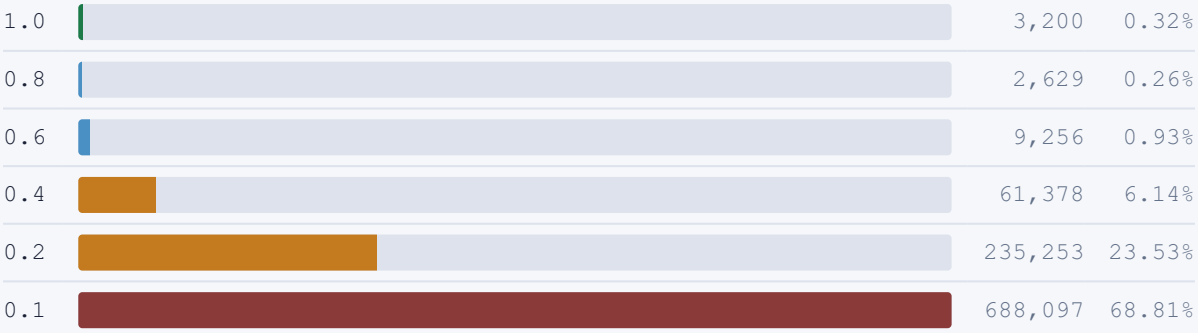
RANK POSITION	MEAN SCORE	TIED WITH GT	SCORED 1.0	STRICTLY BELOW GT
#1 (GT or co-first)	1.000	186/187 (99%)	186/187	0
#2	1.000			0

RANK POSITION	MEAN SCORE	TIED WITH GT	SCORED 1.0	STRICTLY BELOW GT
		186/187 (99%)	187/187 (100%)	
#3	1.000	**186/187 (99%)**	187/187 (100%)	0
#4	0.999	185/187 (99%)	186/187 (99%)	1
#5	0.996	182/187 (97%)	183/187 (98%)	4

Interpretation: The top 3â€“4 positions are almost universally saturated at 1.0. The score-1.0 ceiling is a hard limit of the model's output range, not a sign that these documents are genuinely indistinguishable â€” they span very different embedding distances to GT (from 0.02 to 0.73 within the first distance decile).

NON-GT SCORE DISTRIBUTION

The model uses a strongly quantized output â€” most scores cluster at a handful of discrete values, suggesting the model outputs ordinal assessments rather than a continuous relevance probability:



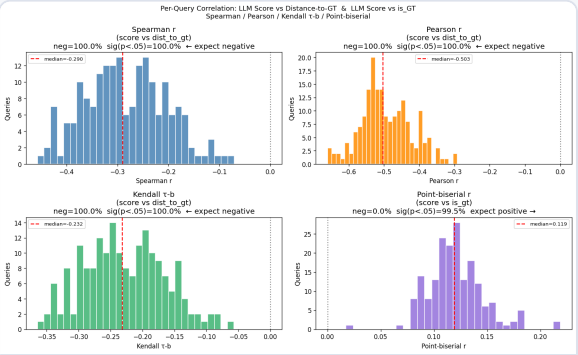
Score quantization: 92.3% of non-GT docs score either 0.1 (irrelevant) or 0.2 (marginal). Only 0.32% of non-GT docs (3,200 of 999,987) score 1.0 â€” these are the false positives that create ties with GT. The model is effectively applying a 6-level ordinal scale, not a continuous score.

Correlation: LLM Score vs. Distance-to-GT

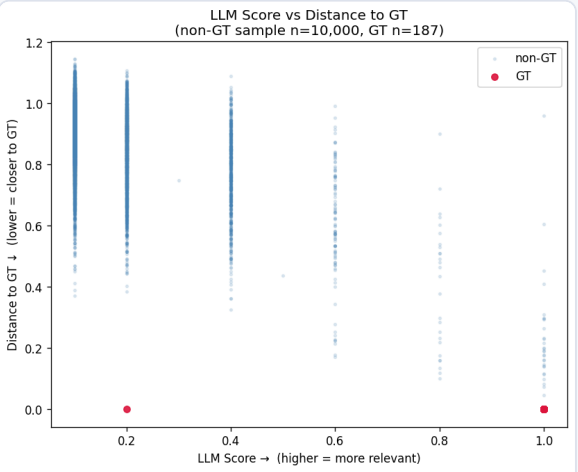
All three correlation measures are consistently negative across all 199 queries â€” LLM score decreases as distance from GT increases.

METRIC	MEDIAN	MEAN	% NEGATIVE	% SIG. (P<0.05)
Spearman r	â^'0.2900	â^'0.2832	100%	100%
Pearson r	â^'0.5031	â^'0.4937	100%	100%
Kendall Ï„-b	â^'0.2322	â^'0.2275	100%	100%
Point-biserial r (vs. is_GT)	+0.1188	+0.1205	0%	99.5%

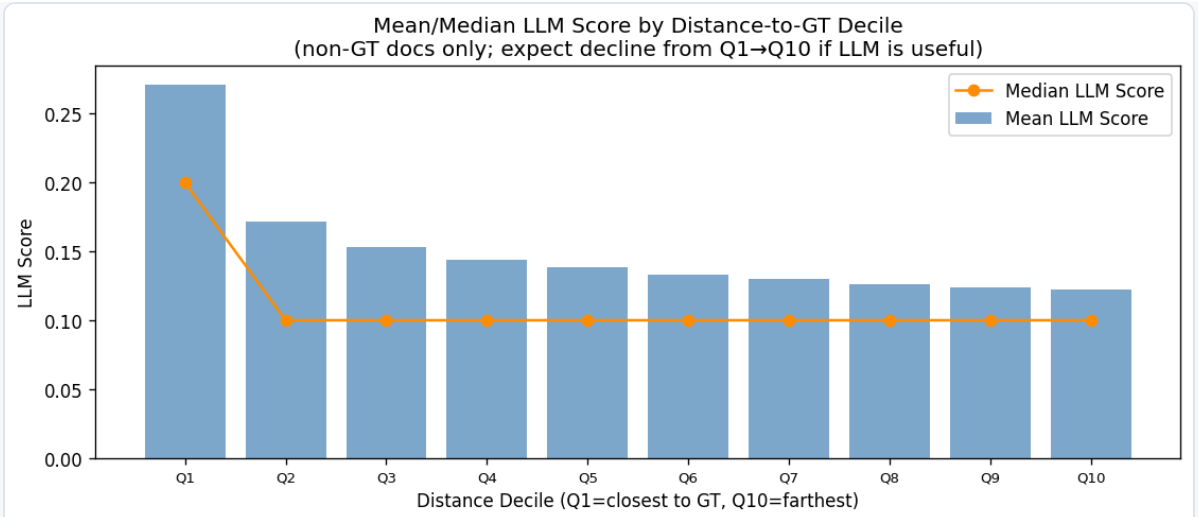
Why Pearson > Spearman: The relationship is not uniformly monotonic â€” GT documents are true outliers (dist â‰ˆ 0, score = 1.0) while the non-GT bulk is flat (dist 0.7â€“1.2, score â‰ˆ 0.1). A linear model captures the GT outlier effect more powerfully than a rank model.



Per-query correlation distributions â€” strictly left-shifted across all metrics. Zero queries show positive correlation.



LLM score vs. dist_to_gt across all 1M pairs. GT cluster (top-left) vs. non-GT mass (lower-right).



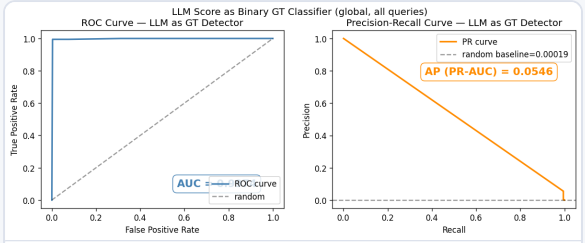
Mean LLM score by decile of dist_to_gt. Score drops monotonically from 0.271 (closest, dist 0.02â€“0.72) to 0.122 (farthest, dist 1.0â€“1.21). Steep drop from decile 0â†’1 driven by GT docs in decile 0 at score 1.0.

DISTANCE QUANTILE DETAIL

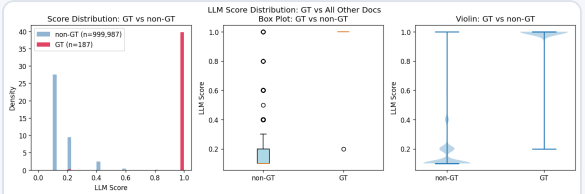
DECILE	DISTANCE RANGE	MEAN LLM SCORE	MEDIAN	COUNT
0 (closest)	(0.022, 0.718]	**0.271**	0.200	99,999
1	(0.718, 0.785]	0.171	0.100	99,999
2	(0.785, 0.827]	0.153	0.100	99,998
3â€“9	(0.827, 1.207]	0.122â€“0.144	0.100	699,993

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GT Detection Performance

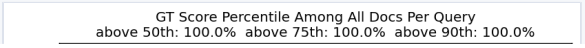


ROC-AUC = 0.997. PR-AUC = 0.055 (292Ã— random baseline â€” low because GT is rare: 1/5,026).



GT mean = 0.996, non-GT mean = 0.151. Near-complete separation with overlap only in the false-positive 1.0 tier.





GT strict rank distribution â€” 186/187 at rank 1. The single outlier (query 29) ranked far down.

GT score percentile within each query's pool. Median = 99.8th percentile â€” GT is almost always in the top 0.2% by score.

MRR & PRECISION SUMMARY

METRIC	VALUE	NOTE
MRR (optimistic)	0.9947	GT shares rank 1 in 99.5% of queries
MRR (tie-corrected)	0.1348	Unbiased estimate assuming uniform tie-breaking
P@1 (optimistic)	99.5%	GT in top score tier
P@1 (tie-corrected)	0.0%	GT is never the unique top-ranked doc
ROC-AUC	0.9974	Near-perfect binary discrimination
GT score percentile (median)	99.8	GT is in top 0.2% by score in every query
GT score = 1.0	186/187	Score ceiling saturates GT

Â§ 7

Conclusions & Recommendations

FINDINGS

1. LLM outperforms the retriever as a GT identifier. Retriever accuracy: 73.8% (138/187). LLM strict-rank-1 accuracy: 99.5% (186/187). More importantly, the LLM fully compensates for every one of the retriever's 49 failures â€” making the combined pipeline 100% accurate on this dataset.

2. Embedding proximity predicts LLM relevance universally. Spearman r = â˜’0.290, Pearson r = â˜’0.503, both negative for 100% of queries with p < 0.05. The correlation is graded across the full distance range, not merely driven by the GT outlier.

3. The LLM is a reliable coarse filter, not a fine ranker. GT is always in the top 0.2% by score, but is never uniquely top-ranked. A median of 18 docs per query receive the same maximum score of 1.0. Tie-corrected MRR = 0.135 is the honest estimate for single-document selection.

4. Score quantization is a structural limitation. The model outputs one of ~6 discrete values (0.1, 0.2, 0.4, 0.6, 0.8, 1.0). 92.3% of non-GT docs score â‰¤ 0.2. Only 0.32% reach 1.0 (false positives). This is a calibration issue â€” the model knows which documents are relevant, but its scoring is too coarse to rank within the relevant tier.

RECOMMENDED NEXT STEPS

Combine retriever + LLM as a two-stage pipeline: retriever to generate a candidate set, LLM to re-score. For the remaining tie-breaking problem (top ~18 docs at score 1.0), options include: (a) a cross-encoder for fine-grained re-ranking within the tied tier, (b) ensemble with embedding similarity as a tiebreaker, or (c) a different scoring prompt that forces ordering rather than binary relevance judgment.

Â§ A

Appendix: Full Metric Tables

CORRELATION SUMMARY

METHOD	MEDIAN	MEAN	% NEGATIVE	% POSITIVE	% SIG.
Spearman r	â^'0.2900	â^'0.2832	100%	0%	100%
Pearson r	â^'0.5031	â^'0.4937	100%	0%	100%
Kendall Ï„-b	â^'0.2322	â^'0.2275	100%	0%	100%
Point-biserial r	+0.1188	+0.1205	0%	100%	99.5%

ROC / PR

METRIC	VALUE
ROC-AUC	0.9974
Average Precision (PR-AUC)	0.0546
Baseline (random) PR	0.000187
PR-AUC / Baseline	292Ã—

MRR & P@K

METRIC	OPTIMISTIC	TIE-CORRECTED
MRR	0.9947	**0.1348**
P@1	99.47%	**0.0%**
P@3 â€“ P@100	99.47%	â€”

Â§ GEOMETRIC GROUP ANALYSIS

Why Centroid-Based Retrieval Groups Consistently Outperform Query-Centered Groups

Three linked experiments investigate the geometric structure of documents in embedding space around the ground-truth (GT) document, revealing why the centroid-based group (G2) captures far more documents than the query-centered group (G1).

GROUP DEFINITIONS

GROUP	CENTER	RADIUS	RULE
G1	Query vector Q	r = dist(Q, GT)	dist(doc, Q) ‰¤ r
G2	Normalised centroid of G3	r = dist(centroid, GT)	dist(doc, centroid) ‰¤ r
G3	â€”	â€”	LLM score = 1.0

Distance = 1 âˆ’ dot(a, b) for L2-normalised embeddings (cosine distance). Dataset: 196 queries Â· 5,035-document pool.

EXPERIMENT 1 Â€” G1 / G2 / G3 GROUP SIZES AND OVERLAPS

G1 MEAN SIZE

3.13

docs Â· median 1.0 Â· density 15.4 docs/radius

G2 MEAN SIZE

9.72

docs Â· median 6.0 Â· density 92.5 docs/radius

G3 MEAN SIZE

17.05

docs with LLM score = 1.0 Â· median 18.0

G2 > G1

77.6%

152 / 196 queries where G2 captures more docs than G1

DIST(CENTROID, GT)

0.098

mean Â· vs dist(Q, GT) = 0.173 â€” centroid 1.77Ã— closer to GT

G2 DENSITY / G1 DENSITY

6.0Ã—

92.5 vs 15.4 docs per unit cosine radius

METRIC	G1	G2	G3
Mean docs	3.13	**9.72**	17.05

METRIC	G1	G2	G3
Median docs	1.0	**6.0**	18.0
GT included (%)	84.7%	85.2%	**94.9%**
Mean centerâ€“GT distance	**0.173**	**0.098**	â€”
Mean density (docs/radius)	**15.4**	**92.5**	â€”
Jaccard G1 âˆ© G2			0.326
Jaccard G2 âˆ© G3			0.336
GT in all three groups (%)			76.0%

EXPERIMENT 2 Â€” HALFPOINT EXPERIMENT (IS THE GT NEIGHBOURHOOD INHERENTLY DENSE?)

For each of 196 queries, 10 random unit vectors are constructed at cosine distance râ,�/2 from GT using Gram-Schmidt orthogonalisation. Each anchor's circle has radius râ,�/2 â€” matching the centroid's typical distance to GT. Result: 1,960 random anchors total.

GROUP	CENTER	MEAN DOCS	MEDIAN	DENSITY
G1	Query Q, radius râ,�	3.18	1.0	15.6
Halfpoint (HP)	Random anchor, radius râ,�/2	**0.82**	1.0	11.0
G2 (reference)	Centroid, radius râ,,	**9.78**	**6.0**	**â€”**

> **Key result:** Random points at the same distance from GT as the centroid capture *fewer* docs (0.82) than even G1 (3.18). Only 7.1% of queries see HP > G1. The GT neighbourhood is **not inherently dense** in arbitrary directions â€” the centroid's advantage requires real document cluster structure.

EXPERIMENT 3 Â€” POOL-DOCUMENT CENTROID (IS THE POOL STRUCTURED AROUND GT?)

For each query, pool documents with dist(doc, GT) < râ,� are identified (closer to GT than the query). Up to 10 are sampled and each used as a circle center with radius = dist(doc, GT) â€” the same convention as G2. Dataset: 159/196 queries have â‰¥1 eligible doc.

GROUP	CENTER	MEAN DOCS	MEDIAN	DENSITY
G1	Query Q, radius râ,€	3.18	1.0	15.6
Pool-doc candidate	Real pool doc, radius=dist(doc,GT)	**10.12**	**6.5**	**59.6**
G2 (true centroid)	G3 centroid, radius râ,,	**9.78**	**6.0**	92.5

Key result: Real pool docs closer to GT than Q capture 10.12 docs on average â€" nearly identical to G2's 9.78. **91.8%** of valid queries see candidate mean > G1. The document pool is structurally concentrated near GT; any real document in the GT neighbourhood works as well as the centroid for raw count.

UNIFIED DENSITY COMPARISON

ANCHOR	MEAN DOCS	MEAN RADIUS	DOCS / RADIUS
Query Q (G1)	3.18	0.173	**15.6**
Random point at râ,€/2 from GT (halfpoint)	0.82	0.087	11.0
Pool doc closer to GT than Q	10.12	~0.070	59.6
G3 centroid (G2)	**9.78**	**0.098**	**92.5**

CONCLUSIONS

(1) G2 outperforms G1 in 77.6% of queries (~3Ã— more docs) because the centroid sits inside the dense document cluster, not in query space.
(2) The GT neighbourhood is not uniformly dense â€" random points yield fewer docs than G1.
(3) The document pool is structurally concentrated around GT â€" any real pool doc near GT captures ~10 docs (91.8% beat G1).
(4) The *queryâ€"document gap* drives the disparity: queries sit in a different embedding region than documents, so using Q as the circle center "wastes" its radius on empty query space.
(5) The centroid is uniquely optimal in density (92.5 vs 59.6 for random pool docs) because it minimises total cluster distance, yielding the smallest radius for the same coverage.

Run: `staged_run_2026-08-03_08-32-45` Â· DB: `experiment_results.db` Â· Model: gcp/google/gemini-2.5-flash-lite via NVIDIA Inference Hub Â· Analysis: analyze_experiment.py, analyze_groups.py, analyze_halfpoint_groups.py, analyze_pooldoc_centroid.py Â· Generated 2026-08-25